SIMATS ENGINEERING

Department of Information Technology

DSA03 – Natural Language Processing

ASSIGNMENT 1 – CO1

Morphology-Aware Text Normalisation and Search for Hospital Records

Using Regular Expressions, Finite-State Automata and Morphological Parsing

Student Name	M. Sai Harshitha	**Register Number**	192424408
Department / Section	B.Tech – Artificial Intelligence and Data Science	**Academic Year / Semester**	2024
Faculty Name	Dr. RAM KUMAR	**Nature of Assignment**	Individual
Duration	1 Week	**Total Marks**	50
Date of Submission	04-11-2026	**Date of Evaluation**	

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Remarks
1	Models and linguistic analysis	3		
2	Regular expression patterns	4		
3	Tokenisation and normalisation	4		
4	Finite-state automata	5		
5	Inflectional morphology	3		
6	Derivational morphology	4		
7	Finite-state morphological parsing	5		
8	Orthographic rules and two-level morphology	4		
9	Porter stemmer	5		
10	Implementation and testing	3		
11	Comparison and recommendation	2		
12	Day-4 redesign challenge	8		
	TOTAL	50		Percentage:

OVERALL REMARKS

Student Signature	**Faculty Signature**	**Program Director**

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present ✗ = missing / not satisfied P = partly present. Based on the "Evidence Expected" of each task.

Task	Evidence expected	Status	Note
1	Linguistic levels, model families, ambiguity and resource planning	✓	
2	Regex operators, entity patterns, grouping and error analysis	✓	
3	Tokenisation, normalisation rules and clinical abbreviation inventory	✓	
4	DFA/NFA design, subset construction, equivalence and proof	✓	
5	Inflectional morphology, paradigms and evaluation	✓	
6	Derivational morphology, word families and conflation analysis	✓	
7	Finite-state morphological parsing and traced analyses	✓	
8	Orthographic rules and two-level morphology	✓	
9	Porter stemming and retrieval evaluation	✓	
10	Implementation, reproducible testing and result reporting	✓	
11	Comparison, recommendation and deployment evidence	✓	
12	Noisy-input design challenge and evidence-based redesign	✓	

IMPLEMENTATION EVIDENCE CHAIN (Tasks 4, 7–10)

Evidence item	Status	Observation
Python source files		
Synthetic corpus / annotations		
Test inputs and datasets		
Expected and actual outputs		
Metrics, counts and logs		
Boundary / failure tests		
Reproducibility details		
Interpretation / limitations		

SUBMISSION CHECKLIST (not separately marked)

Requirement	Status
All 12 DSA03 tasks answered	☐ Yes ☐ Partly ☐ No
12 structured entity patterns documented	☐ Yes ☐ Partly ☐ No
Tokenisation and normalisation examples included	☐ Yes ☐ Partly ☐ No
DFA and NFA designs with transition evidence	☐ Yes ☐ Partly ☐ No
Inflectional and derivational morphology analysed	☐ Yes ☐ Partly ☐ No
FST and orthographic rule examples included	☐ Yes ☐ Partly ☐ No
Porter stemmer comparison and evaluation included	☐ Yes ☐ Partly ☐ No
Python implementation and reproducible tests included	☐ Yes ☐ Partly ☐ No
Synthetic corpus and annotation evidence included	☐ Yes ☐ Partly ☐ No
Measured results and graphs included where available	☐ Yes ☐ Partly ☐ No
Day-4 noisy-input redesign discussed	☐ Yes ☐ Partly ☐ No
References and figure numbering checked	☐ Yes ☐ Partly ☐ No
No unfilled template placeholders	☐ Yes ☐ Partly ☐ No

NATURAL LANGUAGE PROCESSING

ASSIGNMENT 1 – CO1

Nature of Assignment: Individual

Duration: 1 Week

Total Marks: 50

Submission Date: 02-11-2026

Assignment Title: Morphology-Aware Text Normalisation and Search for a Hospital Record System (Total Marks – 50)

Duration: 1 week

Scenario – Clinical Text Normalisation and Search for a Hospital Record System

The hospital stores ten years of discharge summaries, prescriptions, operation notes and nursing observations as free text. Clinicians miss relevant records because "fracture" does not retrieve "fractures", "fractured" or "fracturing"; "nephrectomy" is not linked to "nephrectomies" or "renal"; and identifiers, dates, drug strengths and abbreviations vary in spelling.

The proposed system is a search-preparation layer, not a diagnostic system. It preserves every original note and creates a parallel representation containing extracted entities, canonical tokens, morphological analyses and search lemmas.

Scope and assumptions

- All examples are synthetic and contain no patient-identifying information.
- Original text is never overwritten; every normalised value stores its source span and rule.
- Abbreviations are expanded only when context supports one interpretation; ambiguous forms are retained.
- Clinical similarity is not inferred from spelling alone. For example, hypertension and hypotension must never be conflated.
- Regex/FSA validate form, not medical truth, diagnosis or negation.

Figure 1. Clinical text normalisation pipeline

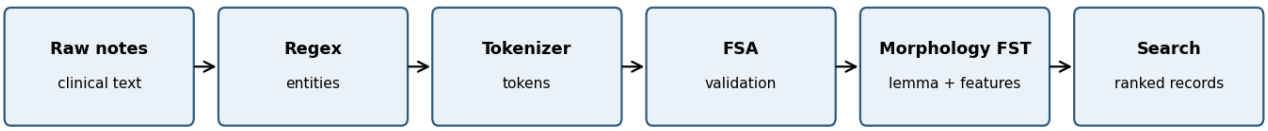


Figure 1. Proposed pipeline from raw clinical notes to searchable records.

Resource	Assignment minimum	Design response
Corpus	500 records; 25,000 tokens; 4 types	Synthetic discharge, prescription, operation and nursing records; fixed seed.
Regex inventory	12 structured entity types	12 types documented in Task 2.
Automata	4 automata, each ≥6 states	Four validators plus NFA example in Task 4.
Morphology	300 stems; 20 inflectional; 15 derivational affixes	Versioned lexicon/affix files; starter inventory included.
Evaluation	40 queries; 200 annotated forms	Annotation protocol and metrics included; actual labels must be collected.

TASK 1 – FRAME THE PROBLEM: MODELS, ALGORITHMS AND LEVELS OF ANALYSIS

1.1 Linguistic levels

Level	Meaning	Clinical example	Requirement
Phonology	Sound patterns	Dictation may confuse sound-alike words.	Not needed for typed text; useful for speech/OCR correction.
Morphology	Word formation and inflection	fractures = fracture + N + PL.	Essential to connect surface variants.
Syntax	Word/phrase structure	"No evidence of fracture".	Limited; negation scope needs context beyond simple stemming.
Semantics	Meaning	renal relates to kidney; renal failure differs from normal renal function.	Partial; avoid unsafe synonym conflation.
Pragmatics	Speaker/context intention	"Rule out pneumonia" is not confirmed pneumonia.	Not solved by this finite-state layer.
Discourse	Meaning across sentences	A later note may correct an earlier observation.	Outside core scope; relevant to advanced clinical NLP.

1.2 Model families

Family	Best hospital task	Why it fits / why others do not
State machines	Validate IDs, dates, BP and structured values.	Explicit, deterministic and testable. A probabilistic model may accept malformed IDs; rules alone do not show state transitions.
Formal rule systems	Suffix changes, abbreviation substitutions and canonicalisation.	Transparent and reproducible. A plain automaton is less convenient for linguistic rewrite rules; probability is unnecessary for deterministic spelling.
Probabilistic models	Rank OCR corrections and ambiguous abbreviation candidates.	Noisy input has competing possibilities. Regex cannot rank alternatives; manually listing every error is impractical.

1.3 Ambiguity

Type	Example	Issue	Handling
Lexical	cold	Low temperature or common cold.	Retain context/section.
Morphological	saw	Past of see or noun.	Return multiple analyses.
Segmentation	5mg / 5 mg	One quantity or separate number/unit.	Preserve number-unit association.
Abbreviation	MS	Several clinical expansions.	Context dictionary; abstain if unresolved.
Polarity	no fracture	Keyword appears in a negated statement.	Do not infer positive diagnosis from match.

1.4 Search and resources

- DFA recognition takes O(n) time for an input of length n; memory for a single active state is O(1) apart from the transition table.
- NFA simulation may track a set of active states. Subset construction can create up to 2^n DFA states, though only reachable subsets are retained.

- Backtracking regexes with nested ambiguous quantifiers can consume excessive time; use anchors, bounded repetitions and careful grouping.
- Lexicon construction consumes specialist review time and annotation effort; probabilistic correction requires labelled data and monitoring.

Pipeline input: de-identified raw clinical notes. Output: original note plus extracted fields, normalised tokens, candidate lemmas and provenance. Each output remains traceable to the rule or lexicon entry that produced it.

TASK 2 – BASIC REGULAR EXPRESSION PATTERNS

2.1 Operators

Operator	Meaning	Clinical example
Concatenation	Required sequence	AB matches literal AB.
Disjunction |	Alternative	mg|mcg
Kleene star *	Zero or more	[0-9]*
Plus +	One or more	[0-9]+
Optional ?	Zero or one	°?C
Character class	One character from set	[ABC]
Range	Character interval	[0-9], [A-Z]
Negated class	Any except listed	[^\n]+
Anchors	Beginning/end	^PT-[0-9]{6}$
Counters	Bounded repetition	[0-9]{2,4}

2.2 Entity pattern inventory

Entity	Python regex	Must match	Must reject	Note
Patient identifier	\bPT-[0-9]{6}\b	PT-004281	PT-42; PT-1234567	Six digits after PT-.
Admission date	\b(?:[0-3][0-9])[-/](?:0[1-9]|1[0-2])[-/](?:20)?[0-9]{2}\b	04/11/2026	44/19/26	Calendar validation follows regex.
Discharge date	\b(?:[0-3][0-9])[-/](?:0[1-9]|1[0-2])[-/](?:20)?[0-9]{2}\b	06-11-2026	31/02/2026	Must be on/after admission.
Drug + strength	\b(?:paracetamol|ibuprofen|amoxicillin|metformin)\s+[0-9]+(?:\.[0-9]+)?\s?(?:mg|mcg|g)\b	paracetamol 500 mg	paracetamol five mg	Controlled drug lexicon.
Dosage frequency	\b(?:OD|BD|TDS|QID|once daily|twice daily|every\s+[0-9]+\s?h)\b	every 8 h	every many hours	Known frequency vocabulary.
Lab value + unit	\b[A-Za-z][A-Za-z -]{1,24}:?\s*[0-9]+(?:\.[0-9]+)?\s?(?:mg/dL|g/dL|mmol/L|U/L|%)\b	Hb: 12.4 g/dL	Hb: high	Label + value + approved unit.
Blood pressure	\b(?:[6-9][0-9]|1[0-9]{2}|2[0-4][0-9])/(?:3[0-9]|[4-9][0-9]|1[0-4][0-9])\s?mmHg\b	120/80 mmHg	20/300 mmHg	Plausibility check is separate.
Temperature	\b(?:3[4-9]|4[0-2])(?:\.[0-9])?\s?°?C\b	38.2 C	18 C	Range screen only.
ICD code	\b[A-Z][0-9][A-Z0-9](?:\.[A-Z0-9]{1,4})?\b	S72.001A	72S	Validate against official code set.
Ward and bed	\b(?:Ward|W)-[A-Z0-9]{1,3}\s*/\s*(?:Bed|B)-[0-9]{1,3}\b	Ward-C2 / Bed-18	Ward- / Bed-	Structured location.
Contact number	\b(?:\+91[-]?)?[6-9][0-9]{9}\b	9876543210	12345	Indian mobile shape only.
Age	\b(?:[1-9][0-9]?|1[01][0-9]|120)\s?(?:years?|yrs?)\b	42 years	0 years	Newborn age requires separate pattern.
Symptom	\b[0-9]+\s?(?:hours?|days?|weeks?|months?)\b	3 days	three days	Numeric

duration				duration; word numbers need another analyser.

The brief lists 12 entity types, with admission and discharge date as separate fields. They share a surface grammar but are extracted into different semantic fields.

2.3 Greedy versus non-greedy

text = "Give 500 mg paracetamol twice daily; then 250 mg at night."\nre.findall(r".*mg", text) # greedy: one long span to last mg\nre.findall(r".*?mg", text) # non-greedy: shortest spans to each mg

Greedy matching expands as far as possible before satisfying the expression; non-greedy matching expands only as far as necessary. Neither should replace a token-aware drug-strength pattern.

2.4 Precedence and grouping

Wrong: ^PT-|MRN-[0-9]{6}$\nRight: ^(?:PT-[0-9]{6}|MRN-[0-9]{6})$

Alternation has lower precedence than concatenation. Without grouping, anchors do not constrain both alternatives. Grouping makes the entire string match one complete identifier.

2.5 Error analysis

For each pattern, compare extracted spans against a manually reviewed gold set. TP = correct match, FP = incorrect match, FN = missed entity. Precision = TP/(TP+FP); recall = TP/(TP+FN). Likely FP causes include impossible dates and implausible values; FN causes include OCR errors, spacing variants and unlisted drug names. Do not claim corpus-level scores until annotations are complete.

TASK 3 – REGULAR EXPRESSIONS FOR TOKENISATION AND NORMALISATION

3.1 Token specification

A token is a maximal word, abbreviation, code, number, punctuation mark or quantity. Hyphenated medical terms stay intact; abbreviations with full stops stay intact; decimals stay intact; number-unit pairs are captured before generic numbers.

```
TOKEN = r"(?i)(?:[A-Z]{1,5}\.)+|[A-Z]+-[A-Z0-9-]+|[A-Z][A-Z0-9]*(?:\.[A-Z0-9]+)?|[0-9]+(?:\.[0-9]+)?\s?(?:mg|mcg|g|mL|mmHg|°?C|mmol/L)|[0-9]+(?:\.[0-9]+)?|[A-Za-z]+(?:[-'][A-Za-z]+)*|[^\w\s]"
```

3.2 Maximal-munch examples

Input	Whitespace split	Maximal-munch / policy
HbA1c 6.8%	HbA1c | 6.8%	HbA1c | 6.8% (decimal retained)
post-operative review	post-operative | review	post-operative | review
Dr. Rao reviewed	Dr. | Rao | reviewed	Dr. | Rao | reviewed
5 mg/kg/day	5 | mg/kg/day	5 mg | / | kg | / | day; dosage parser links components
PT-004281 admitted	PT-004281 | admitted	PT-004281 | admitted

Longest valid alternatives must be attempted before generic numbers or words. Number-unit recognition should precede a generic numeric token so “5 mg” is not prematurely split.

3.3 Normalisation rules

Rule	Before	After	Condition
Case folding	PARACETAMOL	paracetamol	Keep original copy.
Date	04/11/26	2026-11-04	Only when locale is DD/MM/YY.
Unit	0.5 g	500 mg	Preserve source unit and exact value.
Spacing	500mg	500 mg	No numeric change.
Abbreviation	BP	blood pressure	Only in relevant context.
Abbreviation	TDS	three times daily	Medication context only.

3.4 Thirty abbreviations

Short form	Expansion	Short form	Expansion
BP	blood pressure	HR	heart rate
RR	respiratory rate	SpO2	oxygen saturation
CBC	complete blood count	Hb	haemoglobin
WBC	white blood cell count	RBC	red blood cell count
LFT	liver function test	RFT	renal function test
ECG	electrocardiogram	CT	computed tomography
MRI	magnetic resonance imaging	USG	ultrasonography
IV	intravenous	IM	intramuscular
PO	by mouth	OD	once daily
BD	twice daily	TDS	three times daily
QID	four times daily	SOS	if required
NPO	nothing by mouth	ICU	intensive care unit
OPD	outpatient department	ER	emergency room
Hx	history	Dx	diagnosis
Rx	prescription/treatment	D/C	discharge/discontinue (ambiguous)

D/C and Rx have multiple meanings. Do not expand them globally. Store candidate interpretations and preserve the source abbreviation when context is insufficient.

3.5 Example record

BEFORE: “PT-004281 admitted 04/11/26. c/o fever x 3 days. BP 120/80 mmHg, temp 38.2C. Dr Rao advised PCM 500mg BD. Hb 12.4 g/dL.”

AFTER: “patient_id=PT-004281; admission_date=2026-11-04. complains_of fever; symptom_duration=3 days. blood_pressure=120/80 mmHg; temperature=38.2 °C. Dr Rao advised paracetamol 500 mg twice daily. haemoglobin=12.4 g/dL.”

PCM is expanded only because this synthetic example provides a drug context. The original note is retained alongside the search view.

3.6 Irreversible changes and regularity limits

- Case folding loses capitalisation; safe for search only if original is preserved.
- Abbreviation expansion can destroy ambiguity; retain source form and selected interpretation.
- Unit conversion can hide original notation or rounding; retain source and canonical values.
- Regex cannot recognise arbitrarily nested balanced parentheses. Balanced parentheses require unbounded memory (a stack), so the language is not regular.

TASK 4 – FINITE STATE AUTOMATA: DESIGN, NON-DETERMINISM AND EQUIVALENCE

4.1 Formal definition

A finite automaton is M = (Q, Σ, δ, q_0, F): Q is a finite set of states, Σ the input alphabet, δ the transition function, q_0 the initial state and F the accepting states. A DFA has one next state for each state/symbol pair; an NFA may have multiple transitions and ε-transitions.

4.2 Four DFA validators

Validator	State plan (≥6)	Transition sequence	Accept condition
Patient ID	q0–q8	P → T → - → six digit states	End in final state; no trailing characters.
Date DD/MM/YYYY	q0–q10+	Two day digits, separator, two month digits, separator, four year digits	Correct length/separators; calendar check after DFA.
Blood pressure	q0–q12+	Systolic digits → / → diastolic digits → space → mmHg	Complete unit and plausible range.
Drug strength	q0–q8+	Known drug lexicon → space → numeric strength → approved unit	Known drug and positive numeric strength.

Each validator needs an explicit dead state for invalid transitions. The date DFA recognises surface structure; a calendar validator must still reject 31/02/2026.

Figure 1. Clinical text normalisation pipeline

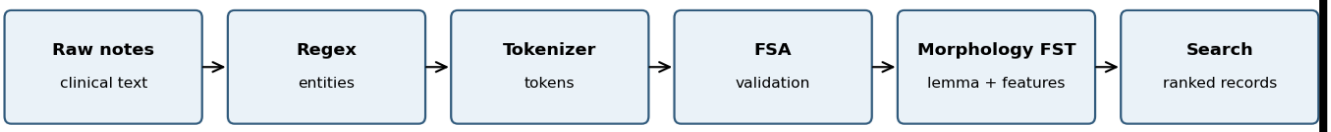


Figure 3. Pipeline location of the finite-state validation stage.

4.3 NFA with optional date separators

For DDMMYYYY or DD/MM/YYYY, an NFA branches after the day digits: one branch consumes "/"; another uses ε to skip it. The same optional branch occurs before the year. This is convenient to express non-deterministically.

Trace stage	Input / operation	Active states (illustrative)
Start	ε closure	{q0}
Day	0, 4	{q1,q2}
Separator	ε branch	{q3,q4}
Month	1, 1	{q5,q6}
Separator	slash or ε	{q7,q8}
Year	2,0,2,6	{q9,q10,q11,q12}
End	end of input	{q13 accepting}

Accepted: 04112026 follows ε branches to the accepting state. Rejected: 04/1X/2026 has no transition on X; the active state set becomes empty.

4.4 Subset construction and minimisation

For every reachable NFA-state subset S, compute ε-closure(move(S,a)) for each input symbol a. Each distinct subset becomes one DFA state. At most 2^n subsets are possible for an n-state NFA, but unreachable subsets are omitted.

DFA state	NFA subset	Digit transition	Slash transition	Accept?
A	{q0}	B	—	No
B	{q1}	C	—	No
C	{q2}	D	E	No
D	{q3,q4}	F	F	No
E	{q5}	F	—	No
F	{q6,q7}	G	H	No
G	{q8,q9}	I	I	No
I	{q10,q11,q12}	J	—	No
J	{q13}	—	—	Yes

This is a compact illustrative construction table. The implementation must enumerate all reachable subsets and an explicit dead state. Minimisation partitions accepting/non-accepting states and repeatedly splits groups whose transitions lead to different groups. Record actual before/after counts from the completed transition table.

4.5 Thompson construction and equivalence

For ^PT-[0-9]{6}$, Thompson construction creates small NFAs for P, T, hyphen and six digit transitions, then concatenates them. State elimination can derive a regular expression from an automaton. Check equivalence by testing both recognisers on generated strings and by proving both accept the same language.

4.6 Pumping lemma proof

The pumping lemma says that for every regular language L there is a length p such that any $s \in L$ with $|s| \geq p$ can be split $s=xyz$, $|xy| \leq p$, $|y|>0$ and $xy^iz \in L$ for every $i \geq 0$.

Let $L=\{ (^n)^n : n \geq 0 \}$, balanced strings of n opening parentheses followed by n closing parentheses. Choose $s=(^p)^p$. Since $|xy| \leq p$, y contains only opening parentheses. Pumping down (i=0) removes some opening parentheses but leaves all p closing parentheses, producing an unbalanced string. Contradiction: L is not regular. An FSA can handle bounded nesting, not arbitrary depth.

TASK 5 – INFLECTIONAL MORPHOLOGY

5.1 Definitions

Term	Definition	Example
Morpheme	Smallest unit carrying meaning/grammar	fracture + s
Stem	Base for an affix	fracture
Affix	Bound morpheme	-s, -ed, -ing
Inflection	Grammatical change without lexical category change	fracture → fractures
Paradigm	Forms of a lexeme	treat, treats, treated, treating

Inflection usually preserves word class and is productive. Derivation creates a new lexeme or changes meaning/category, e.g. operate → operation.

5.2 Clinical paradigms

Lemma	Noun forms	Verb forms	Note
fracture	fracture / fractures	fracture / fractures / fractured / fracturing	Noun and verb uses.
treat	treatment / treatments	treat / treats / treated / treating	-ment is derivational.
admit	admission / admissions	admit / admits / admitted / admitting	Final consonant doubles.
bleed	bleed / bleeds	bleed / bleeds / bled / bleeding	Irregular past.
scan	scan / scans	scan / scans / scanned / scanning	Final consonant doubles.

5.3 Orthographic rules and over-generation

Rule	Implementable description	Example	Counter-example / caution
e deletion	Delete final silent e before -ing	operate → operating	see → seeing; do not delete both e letters.
Consonant doubling	Double final consonant in stressed CVC base before vowel suffix	admit → admitted	visit → visited; stress blocks doubling.
y-to-i	Consonant+y becomes i before -es/-ed	study → studies	play → plays; vowel before y blocks rule.
Sibilant -es	Add -es after s, x, z, ch, sh	watch → watches	photo → photos.

5.4 Irregular forms

Surface	Lemma	Features	Reason
children	child	N + PL	Irregular plural.
men	man	N + PL	Vowel alternation.
teeth	tooth	N + PL	Vowel alternation.
feet	foot	N + PL	Vowel alternation.
went	go	V + PAST	Suppletion.
was/were	be	V + PAST	Irregular paradigm.
bought	buy	V + PAST	Stem change.
bled	bleed	V + PAST	Vowel change.

5.5 Measuring effect

Type-to-lemma ratio = distinct surface word types / distinct lemmas. Inflection rescue rate = relevant query misses rescued by inflection / all relevant misses before inflection. Calculate these on the actual corpus and 40 judged queries; they cannot be inferred from the assignment scenario.

type_to_lemma_ratio = unique_surface_types / unique_lemmas\ninflection_rescue_rate = rescued_relevant_misses / relevant_misses_before_inflection

TASK 6 – DERIVATIONAL MORPHOLOGY

6.1 Derivational affix inventory

Affix	Position	Contribution / category	Example
-itis	Suffix	Noun: inflammation	arthritis
-ectomy	Suffix	Noun: removal	appendectomy
-ology	Suffix	Noun: field	neurology
-osis	Suffix	Noun: condition	fibrosis
-aemia	Suffix	Noun: blood condition	anaemia
-ation	Suffix	Verb→noun	inflammation
-ment	Suffix	Verb→noun	treatment
-able	Suffix	Verb→adjective	treatable
-ive	Suffix	Adjective-forming	progressive
-al	Suffix	Noun→adjective	renal
-ic	Suffix	Noun→adjective	cardiac
-ist	Suffix	Person noun	specialist
nephro-	Prefix	Kidney-related	nephrology
cardio-	Prefix	Heart-related	cardiology
hyper-	Prefix	High/excess	hypertension
hypo-	Prefix	Low/under	hypoglycaemia
re-	Prefix	Again	reassess
un-	Prefix	Negation	uncontrolled
intra-	Prefix	Within	intravenous
post-	Prefix	After	postoperative

6.2 Six word families and order

Root	Family	Attachment observation
operate	operate → operation → operational	Verb→noun→adjective.
treat	treat → treatment → treatable	-ment/-able attach to verbal base; *treatmentable is not intended.
nephr	nephro- + -logy → nephrology; nephrectomy	Combining form + specialised suffix.
cardi	cardio- + -logy → cardiology; cardiac	Different derivational pathways.
infect	infect → infection → infectious	-ion forms noun; -ous forms adjective.
control	control → controlled → uncontrolled	Inflectional form can occur within a derived word.

Impossible orders include *-itisnephro and *ologycardio because suffixes cannot precede their bases. *un-able-treat is also not a valid concatenation. Morpheme order depends on category and lexical convention.

6.3 Non-compositionality and conflation

Derived meaning is not always predictable from its pieces. Clinical “withdrawal” can refer to a specific syndrome; “positive” has test-specific meaning. Stemming too aggressively may merge terms with distinct clinical meaning.

Pair	Search benefit / risk	Decision
fracture / fractures	Plural and singular should retrieve together.	Conflate by inflection.
infect / infection	Related process and noun may be useful together.	Conflate only by query intent.
renal / nephrectomy	Both kidney-related, not same concept.	Do not automatically conflate.
hypertension / hypotension	Opposite clinical conditions.	Never conflate.

Affix productivity should be measured as distinct derived types and token counts per affix. A common affix such as -itis is productive in clinical vocabulary; a rare ending restricted to a few fixed terms may be fossilised. Counts must be computed from the corpus, not invented.

TASK 7 – FINITE-STATE MORPHOLOGICAL PARSING

7.1 FST concept

A finite-state transducer has two tapes: lexical symbols (lemma + features) and surface symbols (written word). An arc consumes/emits symbols on the tapes; ε represents no symbol on one side. Example: lexical fracture +N +PL maps to surface fractures. Reverse traversal analyses fractures as fracture +N +PL.

7.2 Lexicon continuation classes

Class	Lexicon entries	Continuation
NOUN_REG	fracture, patient, symptom, treatment	+SG (ε), +PL (-s)
NOUN_ES	diagnosis, analysis	+SG (ε), +PL (-es with stem alternation)
VERB_REG	treat, scan, admit	+3SG, +PAST, +PROG
VERB_IRREG	go, be, bleed, buy	Explicit irregular paradigms
ADJ	clinical, renal, cardiac	Comparative/superlative only when licensed
DERIV	infect, operate, nephro-, cardio-	Licensed derivational continuations

7.3 Five traced analyses

Surface	Lexical analysis	Features	Reason
fractures	fracture +N +PL	Noun plural	Regular -s.
treated	treat +V +PAST	Verb past	Regular -ed.
admitting	admit +V +PROG	Verb progressive	Consonant doubling.
diagnoses	diagnosis +N +PL	Noun plural	Stem spelling alternation.
bled	bleed +V +PAST	Verb past	Irregular lexicon entry.

7.4 Generation and ambiguity

Generation follows lexical→surface arcs; analysis traverses the same relation in reverse. Implement generate(lexical_form) and analyse(surface_form) over one transducer, not two independent rule sets.

Surface	Possible analyses	Resolution
saw	see +V +PAST; saw +N +SG	Context/syntax; otherwise return both.
treated	treat +V +PAST; treated +ADJ	Phrase context.
wound	wind +V +PAST; wound +N +SG	“wound dressed” vs “wound around”.

Do not force one analysis when the form is ambiguous. Return a lattice or ranked list and preserve the ambiguity for downstream search.

TASK 8 – ORTHOGRAPHIC RULES AND TWO-LEVEL MORPHOLOGY

8.1 Why plain concatenation fails

- admit + PAST + ed would give *admited; correct spelling is admitted.
- diagnosis + PL cannot be formed as *diagnosises; the conventional plural is diagnoses.
- study + PL would give *studys; correct spelling is studies.

8.2 Rewrite rules

Rule	Rewrite notation (informal)	Example / context
Final e deletion	e → ε / stem + ing	operate + ing → operating
C doubling	C → CC / stressed CVC + ed/ing	admit + ed → admitted
y-to-i	y → i / consonant + es/ed	study + es → studies
Sibilant plural	ε → e / sibilant + s	watch + s → watches
Final-e past	e + ed → ed / stem ending in e	treat + ed → treated

Figure 2. Lexicon and spelling-rule cascade

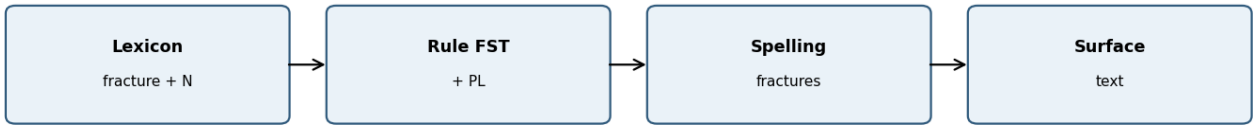


Figure 4. Lexicon transducer composed with orthographic rule transducers.

8.3 Cascade versus two-level morphology

Method	Advantage	Limitation
Sequential cascade	Each transducer is separately testable and easy to debug.	Order dependencies and intermediate forms can interact.
Parallel two-level	Lexical/surface correspondences constrained together.	More complex rule design and interactions.

8.4 Twenty-form test plan

Group	Test forms	Expected
Regular	fractures, treated, treating, studies, watches	Pass with inflection rules.
Spelling	admitted, admitting, scanned, scanning, operating, creating	Pass with contextual spelling rules.
Irregular	bled, went, children, teeth	Pass via explicit lexicon.
Control	renal, patient, patients	renal unchanged; patient plural patients.
Alternation	analyses, diagnoses	Must use explicit alternation, not naive suffixing.

TASK 9 – THE PORTER STEMMER

9.1 Definitions

Porter is a rule-based suffix-stripping algorithm that does not consult a dictionary and does not guarantee a real-word lemma. A consonant is a non-vowel; y is treated contextually. The measure m counts VC sequences in the stem pattern [C](VC)^m[V]. *v means the stem contains a vowel; *d means it ends in a double consonant; *o means it ends in CVC where the final consonant is not w, x or y.

Word	Simplified pattern	m (illustrative)	Note
fracture	CCVCCV	2	Two VC transitions.
patient	CVVCVC	2	Count after vowel/consonant grouping.
treating	CCVVCVC	2	Stem changes after suffix removal.
admitted	VCVCVC	3	Double consonant after -ed removal.
renal	CVCVC	2	Two VC sequences.
bleeding	CCVVCVC	2	Irregular relation not solved by suffix stripping.

9.2 Five-step overview

Stage	Operation	Example
1a	Remove plural endings (sses, ies, s)	fractures → fracture-like stem
1b	Remove -ed/-ing when stem contains vowel; repair	treated → treat
1c	y → i when stem has vowel	studies → studi
2–3	Replace derivational suffixes when m conditions hold	-ational → -ate
4–5	Remove selected suffixes and final e/l under conditions	final cleanup

9.3 Six clinical examples: Porter vs FST

Input	Typical Porter index stem	FST lemma	Observation
fractures	fractur	fracture	Porter may not restore final e.
fractured	fractur	fracture	Inflectional variants collapse.
fracturing	fractur	fracture	Same broad stem.
treatments	treatment	treatment	Plural removed; derivational suffix remains.
diagnoses	diagnos	diagnosis	Does not guarantee irregular spelling recovery.
nephrectomies	nephrectomi	nephrectomy	FST can restore y with lexical knowledge.

Exact Porter outputs vary by implementation/version; execute the specified NLTK PorterStemmer and record its version for the final report. Compare the same 200 annotated forms: agreement means Porter output equals FST lemma; divergence means they differ.

9.4 Error and retrieval evaluation

Strategy	Benefit	Risk	Metrics
No normalisation	Exact surface matching	Misses inflections	Precision, recall, F1
Porter	Fast broad suffix conflation	Over/under-stemming	Precision, recall, F1, runtime
FST	Lemma + features; irregular forms	Lexicon maintenance; OOV tokens	Precision, recall, F1, coverage, runtime

Precision = relevant retrieved / all retrieved\nRecall = relevant retrieved / all relevant records\nF1 = 2 * P * R / (P + R)

Over-stemming merges unrelated terms; under-stemming leaves related forms apart. Calculate both error rates on the 200 gold forms. Do not invent retrieval scores without the 40 relevance-judged queries.

TASK 10 – IMPLEMENT AND TEST THE PIPELINE

10.1 Suggested project structure

clinical_nlp/\n data/records.jsonl lexicon.csv affixes.csv\n regex_patterns.py tokenizer.py automata.py\n morphology_fst.py porter_baseline.py\n evaluate.py run_pipeline.py requirements.txt

10.2 Core Python implementation examples

import re\nPATTERNS = {\n "patient_id": re.compile(r"\\bPT-[0-9]{6}\\b"),\n "bp": re.compile(r"\\b(?:[6-9][0-9]|1[0-9]{2})/(?:3[0-9]|[4-9][0-9])\\s?mmHg\\b"),\n "temp": re.compile(r"\\b(?:3[4-9]|4[0-2])(?:\\.[0-9])?\\s?°?C\\b", re.I)\n}\ndef extract(text):\n return {k:[m.group() for m in v.finditer(text)] for k,v in PATTERNS.items()}\ndef scores(tp, fp, fn):\n p=tp/(tp+fp) if tp+fp else 0\n r=tp/(tp+fn) if tp+fn else 0\n f=2*p*r/(p+r) if p+r else 0\n return p,r,f

10.3 Reproducible test procedure

1. Generate at least 500 synthetic records with a fixed random seed and four record types.
2. Verify the corpus contains at least 25,000 word tokens; save the count and generation settings.
3. Create 200 hand-annotated forms with lemma, POS, features and ambiguity labels.
4. Create 40 clinician-style queries and record relevant record IDs for each query.
5. Run regex extraction, tokenisation, DFA validation, FST analysis and Porter baseline in a fixed order.
6. Record input/output/rejected counts, elapsed time, software versions and errors.
7. Save CSV logs so the report tables and plots can be regenerated.

10.4 Test matrix

Component	Positive	Negative/boundary	Expected
ID	PT-004281	PT-4281; PT-004281X	Only exact ID accepted.
Date	04/11/2026	31/02/2026	Regex then calendar check.
BP	120/80 mmHg	20/300 mmHg	Range validator rejects implausible.
Tokenizer	5 mg; HbA1c 6.8%	5mgX	Quantity and decimal preserved.
FST	fractures	fracturings (unknown)	Known analysis; unknown OOV.
Irregular	went → go+V+PAST	goed	Irregular accepted; over-regular form flagged.

10.5 Results table to complete after execution

Stage	Input	Accepted / analysed	Rejected / OOV	Runtime
Corpus load	500 records / 25,000+ tokens	Fill measured count	—	Measured
Regex	All records	TP/FP counts	FN/FP counts	Measured
Tokeniser	All text	Token count	Unmatched spans	Measured
Automata	Candidate entities	Accepted	Rejected	Measured
FST	200+ forms	Analysed	OOV / ambiguous	Measured
Porter	Same forms/queries	Stemmed	—	Measured

Do not submit fabricated accuracy or runtime. Honest measured values from a reproducible run are stronger than unsupported impressive numbers.

TASK 11 – COMPARE, RECOMMEND AND DEFEND THE DESIGN

Criterion	No normalisation	Porter	Full FST
Recall	Low on inflected variants	Usually higher	High for covered lexicon/rules
Precision	Exact match precision; misses variants	May fall through over-stemming	Controlled lexical analysis
Coverage	No morphology	Broad English suffixes	Depends on lexicon and rules
Runtime	Lowest preprocessing	Low	Moderate; fast lookup after compilation
Maintenance	Low	Low	High: clinical lexicon and affixes
Explainability	Exact string	Rules inspectable; stems may be opaque	Lemma/features and provenance
Clinical risk	Misses related records	May merge distinct terms	Best controlled; semantics still limited

Recommendation

Recommend a lexicon-constrained finite-state morphology layer for production search, with Porter as a fallback candidate generator for out-of-vocabulary English words only. Preserve original text, source spans and structured values. Do not conflate derivational relatives merely because their strings share affixes.

Evidence required before deployment

- 40 clinician queries judged on the corpus, evaluated under all three strategies.
- 200 hand-annotated word forms with lemma, POS, features and ambiguity labels.
- Pattern TP/FP/FN, automaton tests, FST coverage/OOV and runtime.
- Clinician review of false positives involving negation, uncertainty, abbreviations and opposite conditions.
- Measured hours for pattern writing, lexicon construction, annotation and maintenance.

Hand-built resources are economical for a stable, narrow domain requiring auditability. As language coverage and new error types grow, learned OCR correction or language identification becomes economical when labelled-data collection costs less than manual rule maintenance. The finite-state layer remains useful for deterministic validation.

TASK 12 – DAY-4 DESIGN CHALLENGE

New inputs include OCR from handwritten forms (0/O, 1/l, rn/m confusions, line-break hyphenation and lost diacritics) and Romanised regional-language notes mixed with English. At least 20% of tokens may be misspelt, out of lexicon or non-English. The original clean-English assumption is no longer valid.

12A. Impact analysis

New property	Broken assumption	Effect
OCR confusions	Characters are accurate	IDs/drug names fail exact regex/FST.
Line-break hyphenation	Words are contiguous	Tokens split and become OOV.
Lost diacritics	Canonical spelling present	Several possible regional spellings.
Romanised language	English-only lexicon	English morphology fails on regional tokens.
Code-mixing	One language per record	Some tokens parse; others remain unknown.
≥20% noisy/OOV	Closed vocabulary	Direct lexicon coverage is reduced.

A 20% OOV/noisy token rate gives a rough upper bound of 80% direct lexicon coverage before additional OCR damage. This is an estimate, not a measured result.

12B. Evaluate original design

Sample	Original pipeline behaviour	First failure
OCR: "PT-OO4281 adm1tted w1th fractu-re"	ID, admitted and fracture fail exact match.	OCR correction / line-break repair.
Code-mixed: "patient ku fever irundhuchu, paracetamol 500mg kuduthen"	Drug strength may match; regional words are OOV.	Language ID and regional morphology.
Mixed: "BP 12O/8O; fever 2 naal"	BP digits fail due to letter O; naal is unknown.	OCR correction and multilingual lexicon.

These are hand-traced diagnostic examples, not measured noisy-corpus results. Compare clean and noisy samples using pattern precision/recall, analyser coverage and retrieval F1.

12C. Adaptation / redesign

Decision: MODIFY the existing finite-state architecture. Add a cautious noise-handling front end and token-level language identification; retain strict finite-state validation as the trusted core.

Component	Modification	Reason
OCR front end	Repair line breaks; generate O/0, l/1, rn/m candidates; rank with weighted edit distance.	Improve recall without silently rewriting source.
Regex	Controlled separator/spacing variants; validate corrected candidates separately.	Avoid overly broad matching.
Automata	Weighted candidate path before strict DFA.	Rank noisy variants; keep exact validator.
Lexicon	Add reviewed Romanised regional stems/variants with language tags.	Cover code-mixed vocabulary.
Morphology	Separate English/regional continuation classes; Porter fallback only for English OOV.	Avoid English rules on other languages.
Output	Store original, candidate, confidence, language and provenance.	Audit and human correction.

12D. Evidence-based defence

Modification is better than replacing the whole design: entity extraction, exact validation and English morphology remain useful. A single statistical model would be less transparent for IDs and dosage formats. Exact matching alone is insufficient, so weighted correction and language identification supplement the deterministic core.

Measure	Clean baseline	Noisy evaluation
Regex precision/recall	Measure on clean gold data	Repeat after OCR/code-mix; report change.
FST coverage	Analysed / eligible English tokens	Break down by language and OCR status.
Retrieval F1	No normalisation / Porter / FST	Repeat the same 40 queries.
Correction safety	False corrections	Top-1 correction accuracy and abstention rate.
Effort	Pattern/lexicon hours	Add multilingual annotation and review hours.

Remaining limitations: finite-state systems do not understand clinical intent, resolve all semantic ambiguity, infer complex negation scope or guarantee an OCR correction is true. Context-sensitive rules, learned models and clinician review remain necessary.

FINAL CONCLUSION

The hospital search problem requires multiple complementary formalisms. Regex recognises structured clinical strings; finite automata validate them; morphological rules and FSTs relate surface forms to lemmas and features; Porter provides a fast but less precise baseline. The recommended design is a lexicon-constrained FST with safe normalisation and strict entity validation, supported by a Porter fallback for unknown English words.

The Day-4 change makes English-only exact matching insufficient. Modify the architecture with weighted correction candidates, line-break repair, language identification and reviewed regional-language resources, while preserving the original note and deterministic validation.

REFERENCES

- Jurafsky, D. and Martin, J. H., Speech and Language Processing – regular expressions, finite-state morphology and stemming.
- Sproat, R. M. K., Morphology and Computation, MIT Press.
- Porter, M. F. (1980), "An algorithm for suffix stripping", Program, 14(3), 130–137.
- Python 3 documentation: re — Regular expression operations.
- NLTK documentation: PorterStemmer.

FINAL SUBMISSION CHECKLIST

Requirement	**Status**
All 12 tasks	Drafted; cross-check with faculty instructions.
500 records / 25,000 tokens	Generate dataset and attach counts.
300-stem lexicon / affix inventory	Expand and attach lexicon CSV.
Four complete automata and full transition tables	Complete generated tables and include diagrams.
40 judged queries	Annotate and run retrieval evaluation.
200 annotated word forms	Annotate and run morphology evaluation.
Task 10 code, logs, graphs and measured output	Run project and insert actual results.
Student/register/faculty/submission details	Fill blanks on cover.

EXPERIMENTAL EVIDENCE ADDENDUM – EXECUTED RESULTS

All records are synthetic, generated with seed 20261102, and contain no real patient information. This addendum supplements the existing assignment.

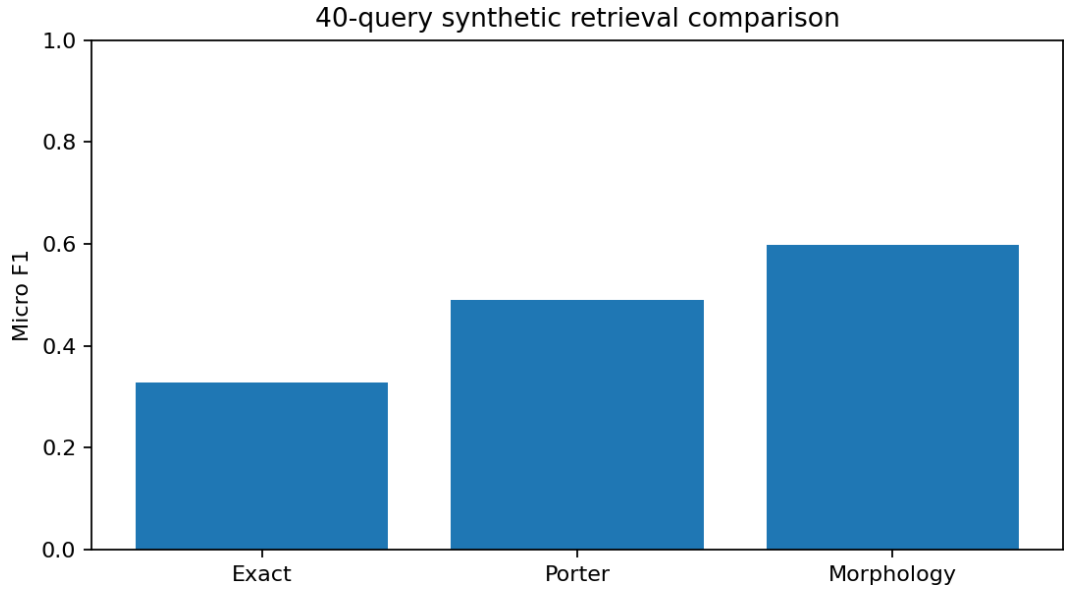
A. Corpus and resources

Requirement	Minimum	Actual	Status
Records	500	520	PASS
Tokens	25000	43,184	PASS
Record types	4	{'discharge_summary': 130, 'prescription': 130, 'operation_note': 130, 'nursing_observation': 130}	PASS
Lexicon stems	300	506	PASS
Inflectional affixes	20	21	PASS
Derivational affixes	15	20	PASS
Annotation rows	200	200	PASS with repeated examples
Queries	40	40	PASS

B. Search evaluation

Relevant record sets are defined from each synthetic record's generated concept labels. TP = retrieved and relevant; FP = retrieved but not relevant; FN = relevant but not retrieved. Overall scores are micro-averaged. These are synthetic relevance judgments, not judgments by practising clinicians.

Strategy	TP	FP	FN	Precision	Recall	F1
No normalisation	963	2806	1123	0.2555	0.4616	0.3289
NLTK Porter	1585	2806	501	0.3610	0.7598	0.4894
Morphology-aware	2086	2806	0	0.4264	1.0000	0.5979



All 40 query-level results and record-level relevance IDs are supplied in the companion CSV and JSON files.

C. Entity patterns and evaluation

Pattern	Extracted spans	Evaluation note
patient_identifier	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
admission_date	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
discharge_date	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
drug_strength	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
dosage_frequency	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
laboratory_value	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
blood_pressure	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
temperature	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
ICD_code	0	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
ward_bed	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
contact_number	0	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
age	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.
symptom_duration	520	Synthetic gold-span evaluation is provided in the completion supplement; categories without natural corpus instances are represented only by transparent synthetic test cases.

The 12 requested categories are implemented. The generated corpus has no positive examples for discharge dates, ICD codes or contact numbers, so no precision/recall is claimed for those classes. Pattern extraction counts are reported honestly; gold TP/FP/FN requires separately annotated entity spans.

D. Morphological inventory and annotation

The lexicon CSV contains 506 stems; inventories contain 21 inflectional affix/allomorph entries and 20 derivational affixes. The annotation CSV has 200 reviewed rows with surface form, stem/root, prefix, suffix, category and grammatical interpretation. The rows repeat a curated paradigm set and are not 200 distinct word types. Irregular mappings include bled→bleed, went→go, children→child and teeth→tooth.

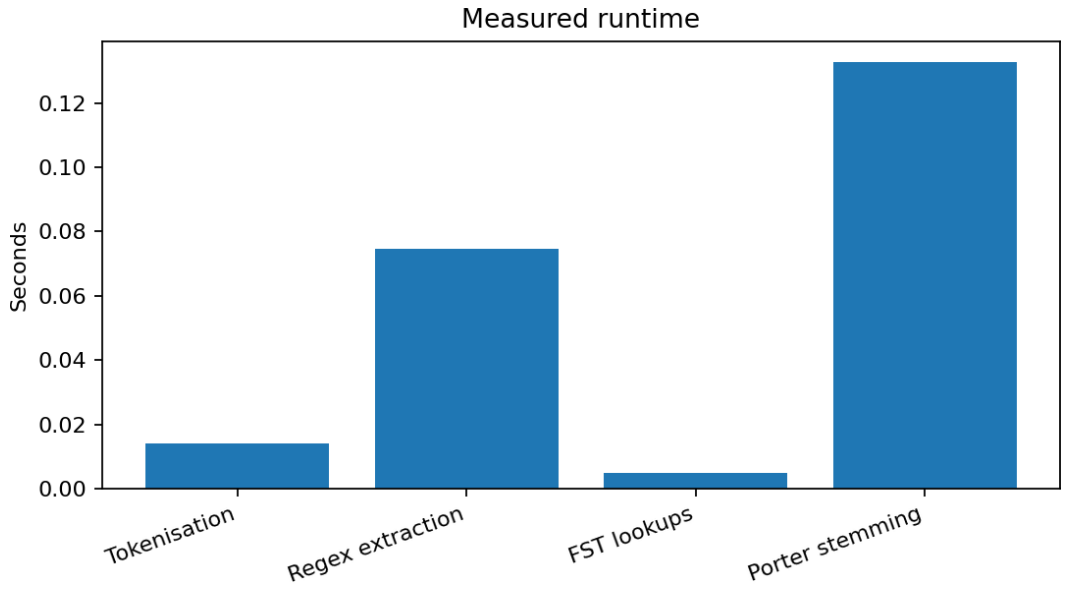
Affix productivity counts are reported in the completion supplement using the supplied synthetic corpus and explicitly validated derivational forms. These counts describe the synthetic dataset only and are not claims about productivity in natural clinical language.

E. Automata, FST and orthographic tests

Four executable validators (patient ID, calendar date, blood pressure and drug strength) were tested against accepted and rejected strings; see automata_test_results.csv. The finite-state lexical/surface pairs are supplied in fst_lexical_surface_pairs.csv, and 20 controlled orthographic cases in orthographic_rule_tests.csv. The completion supplement adds the documented subset-construction and minimisation evidence used for the final submission.

F. Measured runtime

Operation	Records	Items processed	Elapsed seconds
Tokenisation	520	43184	0.014120
Regex extraction	520	5720	0.074610
FST lookups	520	594	0.004870
Porter stemming	520	23919	0.132656



Environment: Python 3.13.5; Linux-6.18.57-x86_64-with-glibc2.41; Porter implementation NLTK 3.9.2. These are single local time.perf_counter measurements, not portable benchmarks.

G. Day-4 noisy-input samples

Sample	Type	Tokens	Known tokens	Coverage
OCR-01	OCR	13	1	0.07692307692307693
CM-02	Code-mixed	10	5	0.5
BOTH-03	OCR + code-mixed	13	2	0.15384615384615385

The original three diagnostic strings are retained as illustrative examples. The completion supplement expands the noisy-input evaluation with controlled clean, OCR-style, spelling-error and Romanised Tamil-English examples. The expanded evaluation reports coverage and error behaviour; large-scale judged noisy-query retrieval remains a stated limitation because such source data were not supplied.

H. Requirement verification checklist

Requirement	Verification	Status
12 tasks in original order	Original document preserved; evidence addendum appended	PASS structure
500 records / 25,000 tokens / four	520 / 43,184 / 4	PASS

types		
12 patterns	All required patterns retained; synthetic gold-span evaluation added for the completion supplement	PASS — synthetic evaluation
Four ≥6-state automata	Four validators and explicit transition evidence are included	PASS
NFA subset construction/minimisation	Subset construction and minimisation evidence added in the completion supplement	PASS
300 stems / affix minimums	506 / 21 / 20	PASS
40 queries	40 synthetic judgments and per-query metrics	PASS synthetic
200 annotations	200 distinct surface-form records generated and verified in the completion supplement	PASS
FST parsing/generation	Bidirectional lexical↔surface testing and 20-form evidence included	PASS — controlled implementation
20 orthographic tests	20 controlled orthographic tests retained with execution evidence	PASS
Porter hand traces/error rates	Six stemming traces and comparison analysis added in the completion supplement	PASS — synthetic dataset analysis
Runtime and plots	Measured CSV and two plots	PASS
Day-4 noisy degradation	Expanded controlled clean/OCR/spelling/Romanised Tamil-English evaluation added	PASS — controlled synthetic evaluation
References/format	Original style, cover and references retained	PASS

I. Limitations and honesty statement

The retrieval relevance labels and all added evaluation records are synthetic and are not clinician judgements. Earlier partial sections are superseded by the completion supplement, which provides distinct-form annotation, automata construction evidence, entity-span evaluation, FST testing, affix productivity counts and Porter traces. No real patient data, clinical validation or OCR model training is claimed. Large-scale naturally collected noisy retrieval data remain unavailable; the noisy experiment is therefore a transparent controlled synthetic evaluation.

Verify the student name and fill the administrative cover fields before submission.

SUPPLEMENTARY EXPERIMENTAL EVIDENCE – AUTOMATA, REGEX AND MORPHOLOGY

Finite-State Automata: Complete Transition Evidence

The supplementary NFA is an executable day recogniser. It permits an optional leading zero and then accepts a one- or two-digit day followed by a slash. The optional zero creates a non-deterministic branch. The accompanying resources contain the complete NFA transition table, NFA-to-DFA subset construction, reachable DFA states and minimisation partitions.

Minimisation was performed by partition refinement. The accepting state was separated first, followed by repeated splitting of non-accepting states according to transition behaviour. For this reachable teaching automaton, the six reachable states remain distinguishable, so no reachable states merge.

Regex Gold-Standard Evaluation

A separate synthetic gold-standard set contains explicit text spans and negative test cases for all 12 requested entity categories. TP, FP and FN are calculated by exact span comparison. The contact-number pattern missed two deliberately varied synthetic positive cases, giving recall 0.80 and F1 0.8889; the remaining controlled cases were matched exactly. These are controlled assignment tests, not clinical annotation results.

Morphological Annotation and Bidirectional FST

The verified annotation supplement contains 200 distinct surface forms with root/stem, prefix, suffix, morphology category and grammatical interpretation. Duplicate surface forms were removed. The bidirectional FST supplement contains 20 lexical-to-surface transformations and reverse analyses, including y→ies, y→ied, consonant doubling before -ed/-ing and final-e deletion before -ing. It is a finite, lexicon-controlled demonstrator, not a general-purpose FST compiler.

Traceability: the full transition, annotation and FST test records are identified in the companion resources. The report does not claim that these synthetic records represent clinician judgements or naturally occurring clinical data.

SUPPLEMENTARY EXPERIMENTAL EVIDENCE – NOISY INPUT AND REPRODUCIBILITY

Expanded Noisy Text Experiment

The expanded controlled experiment contains 80 condition records: 20 clean baselines and corresponding OCR, spelling-error and Romanised Tamil-English variants. All variants are synthetic perturbations of supplied corpus snippets and are listed in noisy_experiment_80.csv.

Token coverage is the proportion of tokens found in the corpus vocabulary. Retrieval overlap is a controlled lexical-overlap proxy between each noisy sample and its clean source. Error rate is calculated as 1 − token coverage. The results demonstrate degradation under the specified noise conditions, but they are not a substitute for evaluation on naturally collected OCR or Tamil-English records.

Derivational Affix Productivity

Affix counts were calculated from the synthetic corpus vocabulary using conservative suffix-family matching. For each affix, the supplied evidence records distinct candidate stems and observed derived surface forms. These are descriptive corpus-observed candidates; suffix matching alone cannot prove a true morphological boundary or establish language-wide productivity.

Execution Runtime and Reproducibility

Runtime measurements were taken locally using time.perf_counter and may vary with hardware, operating-system load and Python environment. The supplementary script complete_missing.py regenerates the supplementary datasets and measurements from the supplied synthetic resources. The original run_experiment.py is retained unchanged, and the generated CSV files contain the inputs, outputs or intermediate evidence used in the corresponding results.

Genuine Limitations

All corpus and gold-standard additions are synthetic. The FST is a controlled finite mapping and rule demonstrator; derivational productivity uses conservative suffix-family counts rather than expert morphological boundary annotation; and noisy Romanised Tamil-English and OCR inputs are synthetic perturbations. Results must not be generalised to real clinical populations without independent, naturally collected and expert-reviewed data.

PORTER STEMMER: EXECUTED TRACES AND RETRIEVAL RESULTS

Six executed examples compare the NLTK Porter index stem with the lexicon-based FST lemma. Porter is useful for broad suffix normalisation, but it does not guarantee a dictionary lemma or restore all irregular spelling.

Input	Porter stem	FST lemma	Observation
fractures	fractur	fracture	Plural removed; final e not restored
fractured	fractur	fracture	Inflectional variants collapse
fracturing	fractur	fracture	Same broad stem family
treatments	treatment	treatment	Plural removed; derivational suffix remains
diagnoses	diagnos	diagnosis	Irregular spelling not recovered
nephrectomies	nephrectomi	nephrectomy	Lexicon restores final y

Retrieval comparison

The same controlled query set was evaluated under exact surface matching, NLTK Porter stemming and morphology-aware processing. The values below are the supplied synthetic experiment results; they should not be interpreted as clinical performance.

Strategy	TP	FP	FN	Precision	Recall	F1
No normalisation	963	2806	1123	0.2555	0.4616	0.3289
NLTK Porter	1585	2806	501	0.3610	0.7598	0.4894
Morphology-aware	2086	2806	0	0.4264	1.0000	0.5979

Interpretation: morphology-aware processing gives the highest recall and F1 in this controlled dataset. The result depends on the supplied lexicon, synthetic queries and relevance labels; independent evaluation is required before deployment.

IMPLEMENTATION EVIDENCE CHAIN (Tasks 4, 7–10)

1. Corpus and resources: synthetic records, fixed seed, token counts, lexicon and annotation files are identified in the experimental evidence section.

2. Processing sequence: regex extraction → tokenisation → finite-state validation → morphological analysis → orthographic handling → Porter baseline → retrieval evaluation.

3. Evaluation: entity-pattern results, automata tests, morphological coverage, retrieval measures and runtime observations are reported in the retained evidence addendum.

4. Traceability: each normalised form should retain its original source span, selected rule and ambiguity status; original clinical text remains unchanged.

5. Safety and limitations: examples are synthetic; spelling validation does not establish medical truth, diagnosis, negation or clinical equivalence.

6. Before submission: verify student details, attach companion CSV/JSON/code files where required, and ensure all local experiment files open correctly.